Communication Structure

Tello Agent JSON Messages

Purpose	Destination IP	Message Type	Key Fields
Status Information	192.168.1.255	status_information	least_cost_mission: "None" ip_address: drone_ip_address status: "WAITING" traffic: "broadcast"
Drone Mission Information	192.168.1.255	drone_mission_information	name: "" least_cost_mission: least_cost_mission ip_address: drone_ip_address status: "WAITING" traffic: "broadcast"
Mission Completed	192.168.1.255	mission_completed	tello_name: "TELLO-B5997E"
QR Code Check	192.168.1.255	qr_code_check	tello_name: "TELLO-B5997E"
Battery Status	192.168.1.255	battery_status	tello_name: "TELLO-B5997E" battery_level: "75%"
GPS Location Update	192.168.1.255	gps_update	tello_name: "TELLO-B5997E" latitude: "37.7749" longitude: "-122.4194"

Ground Control JSON Messages

Purpose	Destination IP	Message Type	Key Fields
Start Mission Information	ip_address	ground_start_mission_information	mission_matrix: self.mission_matrix command: "start" traffic: "broadcast"
Information Wait Command	ip_address	info	mission_matrix: self.mission_matrix command: "wait"
Mission Status Update	ip_address	mission_status	mission_id: "12345" status: "in_progress"
Emergency Stop Command	ip_address	emergency_stop	mission_id: "12345" reason: "obstacle_detected"

Purpose	Destination IP	Message Type	Key Fields
Data Request	ip_address	data_request	request_type: "sensor_data" time_range: "last_5_minutes"
Configuration Update	ip_address	config_update	config_id: "67890" new_settings: { "max_speed": "10m/s", "altitude_limit": "30m" }